THE LABOR PARADOX

The Uncomfortable Coexistence of Job Destruction and Labor Shortages

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EXECUTIVE SUMMARY

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RESEARCH BACKGROUND & MOTIVATION

Modern society faces two simultaneous and seemingly contradictory megatrends: the rapid advancement of AI and automation, widely predicted to eliminate millions of jobs, and ultra-low fertility rates combined with rapid population aging, projected to cause severe labor shortages in the coming decades. Jobs are disappearing—yet the workforce is shrinking even faster. This report offers the first systematic, integrated analysis of how these two forces interact to reshape the future of work.

South Korea exemplifies this paradox in extreme form: the world's lowest total fertility rate (0.8 in 2025) alongside the world's highest industrial robot density (1,220 robots per 10,000 manufacturing workers, IFR 2025).

The core conclusion is that the 'Labor Paradox' is not a true paradox but a transition friction—a temporal, spatial, and skills mismatch between jobs being destroyed and jobs being created. History shows technological progress increases long-run employment (global labor productivity ×8 from 1950–2020 while employment ×3), yet the AI revolution differs in speed, scope, and depth, simultaneously replacing both manual and knowledge work.

CHAPTER HIGHLIGHTS

Chapter	Key Insight
Ch. 2 Historical Context	Four industrial revolutions analyzed through the lens of core technology, energy transition, and employment structure. Each revolution destroyed existing jobs while creating new ones—always with painful transition friction.
Ch. 3 Technology & Labor	White-collar and high-skilled knowledge roles are now vulnerable. AI reduces time spent on routine cognitive tasks by ~81% (author's personal productivity experiment). Gig economy covers 150–450M workers globally but lacks safety-net coverage.
Ch. 4 Future Scenarios	Two pathways to 2070: a positive scenario (automation test-bed, 4-day week, UBI, zero-marginal-cost society) vs. a negative scenario (mass unemployment, platform monopoly, extreme polarization, AI authoritarianism). The divergence point is redistribution capacity.
Ch. 5 Roles & Responsibilities	Governments must reform education (AI literacy, lifelong learning), strengthen safety nets (flexicurity, UBI pilots), and regulate AI (EU AI Act model). Corporations must pursue CSV, Just Transition, and ESG. Individuals must invest in AI literacy, creativity, and civic engagement.

Ch. 6 Policy Roadmap	Short-term (2025–2030): mandatory AI literacy, flexicurity, risk-based AI regulation, UBI pilots. Mid-term (2030–2040): 4-day week legislation, UBI expansion, acceleration towards 2050 Carbon Neutrality. Long-term (2040–2070): full UBI, redefinition of labor as self-realization.

CORE POLICY RECOMMENDATIONS

- Mandatory AI literacy from elementary school across all OECD nations
- Danish-style flexicurity extended to platform and gig workers
- Risk-based AI regulation framework modeled on the EU AI Act (2024)
- Regional UBI pilot programs as precursor to full implementation
- Legislative pathway toward a 4-day / 35-hour workweek by 2030
- Robot tax and wealth tax to fund redistribution as automation accelerates

Author's Note: This research project was conducted by a pre-collegiate researcher using an integrated approach combining traditional research with generative AI tools (Gemini, Claude, Perplexity). All key statistics and citations were manually verified. An embedded productivity experiment demonstrated an 81% reduction in knowledge-work time through AI assistance—illustrating the very phenomenon this report analyzes.

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1. Introduction

1.1 Research Background and Motivation

Modern society stands at the intersection of two unprecedented megatrends. On one hand, the rapid advancement of artificial intelligence (AI) and automation technologies is widely predicted to eliminate large numbers of jobs. On the other hand, ultra-low fertility rates and rapid population aging are projected to cause severe labor shortages in the coming decades. Jobs are disappearing—yet the number of people available to perform work is shrinking even faster.

How can we explain this seemingly contradictory situation? If AI and robots can truly replace human labor on a massive scale, why do governments, companies, and societies continue to worry about not having enough workers? This question goes beyond mere curiosity; it represents a core puzzle that will fundamentally reshape the future structure of work and society.

Despite reviewing numerous academic papers, global reports (WEF, OECD, ILO, UN, McKinsey, Goldman Sachs, etc.), and expert analyses, I found that most studies treat technological job displacement and demographic decline as separate issues. Very few attempt a systematic, integrated examination of how these two forces interact and jointly transform the labor market. This intellectual gap became the primary motivation for the present report.

The Reality of Job Destruction

The World Economic Forum's Future of Jobs Report 2023 estimates that, between 2023 and 2027, approximately 83 million jobs will disappear globally while only 69 million new jobs will be created—resulting in a net loss of 14 million positions (roughly 2% of global employment).

The Paradox of Labor Shortages

At the same time, 16 OECD countries are experiencing ultra-low fertility (TFR ≤ 1.3). South Korea recorded a total fertility rate of 0.8 in 2025. According to the United Nations (2024), the global share of the working-age population (ages 15–64) is projected to decline from 65.0% in 2023 to 59.6% by 2100.

Personal Motivation

The author's grandparents operated a Hanbok store in a Busan traditional market in South Korea for their entire working lives. Yet the explosive growth of e-commerce platforms such as Amazon and Coupang ultimately rendered their business unsustainable. This is not an isolated

family story—thousands of neighborhood shops and traditional markets across the country have faced the same fate. Witnessing how quickly technological change can dismantle an entire generation's livelihood compelled me to ask: What competencies must my generation develop to navigate this rapidly shifting world?

1.2 Significance of the Study

This report aims to deliver the following value:

- An integrated analytical framework that simultaneously addresses two megatrends: technological disruption and demographic structural change
- A long-term historical perspective on how technological revolutions have continuously reshaped labor markets
- Concrete, actionable roles and responsibilities for governments, corporations, and individuals
- A youth-centered exploration of future occupational environments and the capabilities required to thrive

Unlike most existing discussions that examine job destruction by AI and fertility decline in isolation, this study focuses on their mutual interaction and combined impact on the future.

2. Industrial Revolutions and the History of Labor

2.1 The Flow of the Industrial Revolutions

In human history, industrial revolutions represent fundamental transformations not only in technology but across economy, society, and culture as a whole. Each revolution is characterized by innovations in core technologies, shifts in energy sources, and changes in production methods, inevitably leading to the restructuring of labor markets and employment structures.

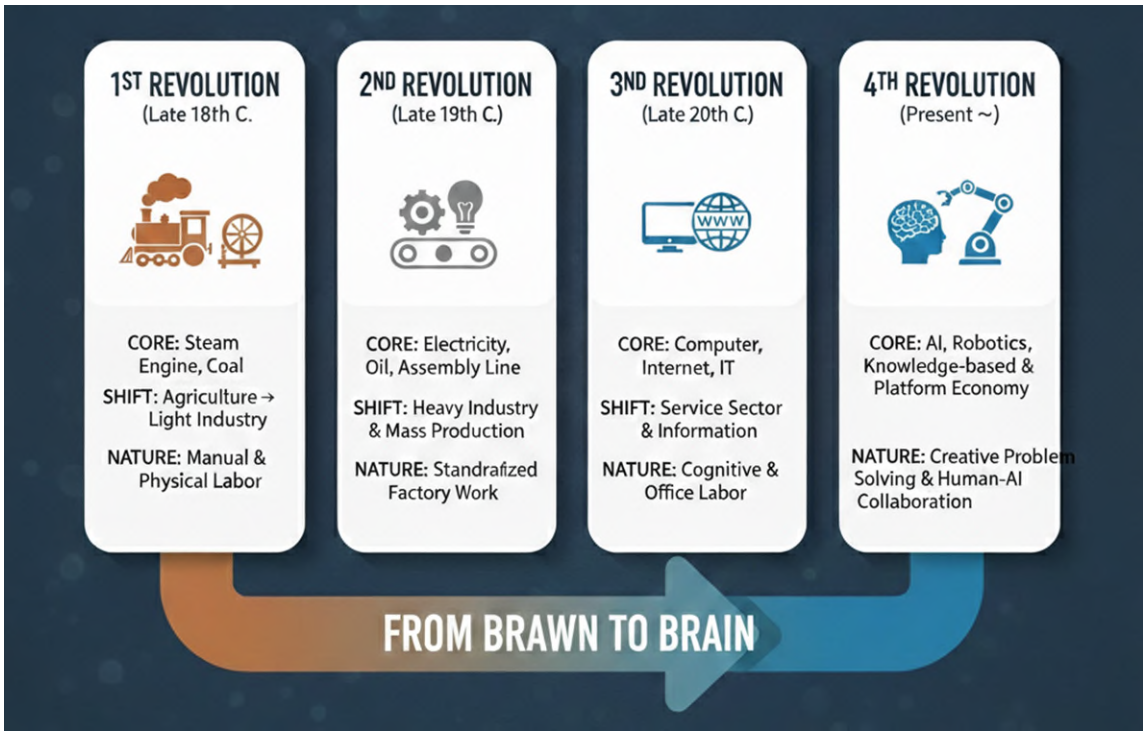


Figure 2-1: Evolution of Labor Structure

1st Industrial Revolution (1760–1840): Steam Engine and Mechanization

The commercialization of James Watt's improved steam engine in 1784 marked a turning point in human history. Mechanization began in the textile industry and spread throughout manufacturing. George Stephenson's steam locomotive in 1825 enabled mass transportation and dramatically expanded market reach.

The socioeconomic impact was dramatic. Large-scale rural-to-urban migration occurred, and industrial cities such as Manchester and Birmingham grew rapidly. The aristocratic feudal order collapsed, giving rise to the capitalist and working classes. However, early factory labor conditions were extremely harsh: 14–16 hour workdays, dangerous environments, low wages, and widespread child labor were common.

The labor market fundamentally changed. In England, the proportion of agricultural workers fell from about 75% in 1750 to about 22% in 1850, while manufacturing and mining rose to around 24%. Traditional artisans lost their livelihoods competing against machines; skilled weavers in particular faced wage declines and unemployment due to spinning jennies and power looms.

Skilled weavers resisted machine introduction through the Luddite movement (1811–1817). The British government suppressed it with legislation against machine-breaking and the deployment

of 14,000 troops; 17 ringleaders were hanged. Yet their resistance later became the seed for labor rights legislation.

2nd Industrial Revolution (1870–1914): Electricity and Mass Production

From the 1870s, the second industrial revolution placed "electricity" in human hands and ushered in the age of mass production. Thomas Edison's practical incandescent bulb in 1879 banished night; Carl Benz's internal combustion automobile in 1885 and the Wright brothers' powered flight in 1903 redefined time, space, and mobility.

The true game-changer, however, was Henry Ford's introduction of the moving assembly line at his Detroit plant in 1913. Model T production time dropped from 12 hours to 93 minutes, and price fell from $825 in 1909 to $260 in 1925—transforming automobiles from luxury items into everyday transportation for ordinary people. Combined with Frederick Taylor's scientific management, work became highly standardized and divided.

This brought dual impacts on the labor market. Workers at the conveyor belt were no longer "skilled artisans." Skill level became less important, enabling mass employment of unskilled labor, while simultaneously creating explosive demand for high-skilled personnel: electrical engineers, mechanical designers, chemists, production managers, and a surge in clerical workers. In other words, the second revolution did not merely "eliminate jobs"—it simultaneously destroyed skilled crafts and gave birth to a new professional class.

3rd Industrial Revolution (1960–2000): Computers and Informatization

From the late 1960s, the third revolution—centered on computers, semiconductors, and the internet—fundamentally transformed economy and society. The transistor (1947) ended the vacuum tube era; the integrated circuit (1958) made computers smaller and more powerful; the World Wide Web (1991) enabled global free connection of information.

Manufacturing jobs declined while service sector employment exploded. In the US, service sector employment rose from about 55% in 1950 to about 78% in 2000. A new class of knowledge workers emerged—programmers, data analysts, systems engineers—while repetitive clerical roles such as typists, switchboard operators, and cashiers rapidly disappeared due to automation. Labor shifted from "hands" to "minds," and information processing and creativity became the keys to success.

4th Industrial Revolution (2016–present): AI, IoT, Big Data, and Robotics

In 2016, Klaus Schwab of the World Economic Forum described the fourth industrial revolution

as an unprecedented fusion where "physical, digital, and biological boundaries blur." Key keywords are fusion, hyper-connectivity, and super-intelligence.

AI now invades domains once believed exclusive to humans: image recognition, natural language understanding, and complex decision-making. Especially since ChatGPT's appearance in 2022 demonstrated that generative AI can perform creative, coding, and analytical high-level knowledge work, shockwaves have spread worldwide. IoT connects everything to generate real-time data; big data converts that vast information into real value.

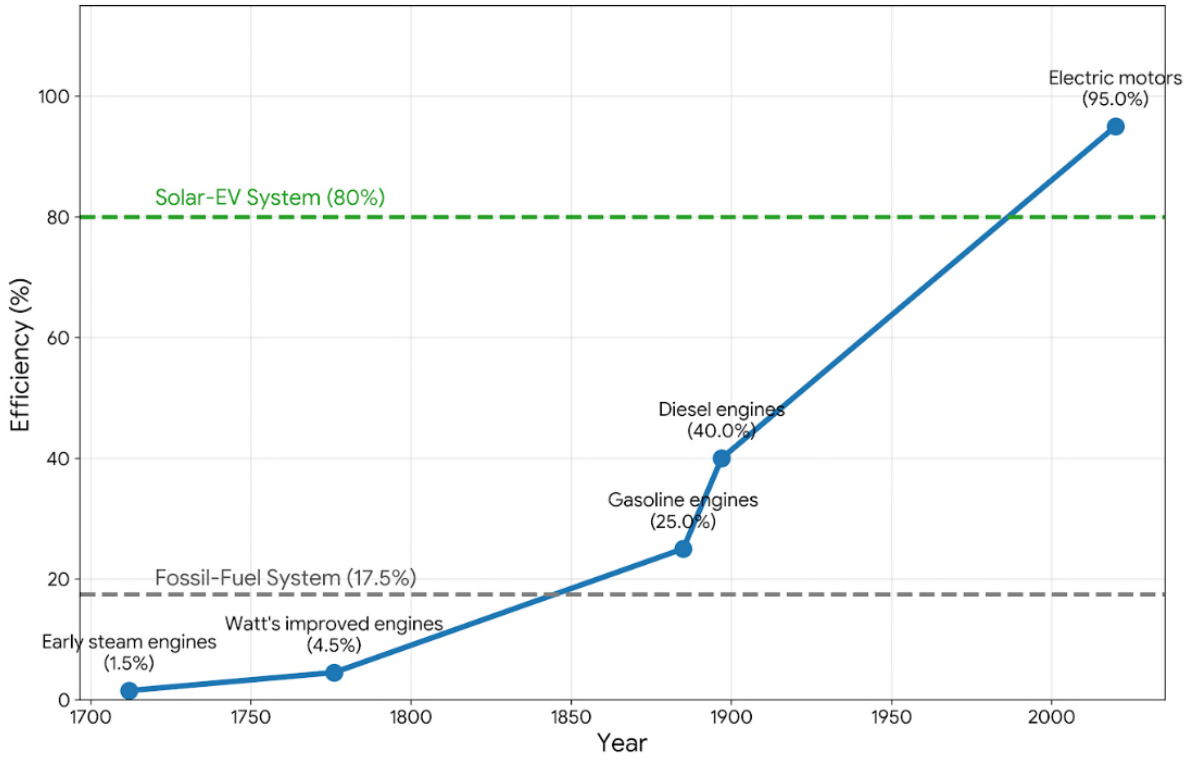
Interestingly, South Korea simultaneously holds the world's lowest total fertility rate (0.8 in 2025) and the highest robot density (1,220 robots per 10,000 manufacturing workers, IFR World Robotics 2025). This shows an extreme "compressed phenomenon" of population decline and automation progressing simultaneously.

The labor market faces unprecedented upheaval. According to WEF Future of Jobs Report 2025 and Goldman Sachs analyses, 25–30% of global work hours may be automated in the next 5–10 years, with generative AI potentially affecting approximately 46% of white-collar tasks. Particularly vulnerable areas include transportation/logistics, manufacturing, clerical/administrative work, and customer service. Conversely, new opportunities are emerging rapidly: AI/machine learning specialists, data scientists, sustainability experts, robotics engineers, cybersecurity professionals, drone pilots, and virtual human creators. The core challenge is narrowing the skills mismatch between disappearing and emerging jobs.

Table 2-1: Key Technologies and Employment Changes by Industrial Revolution

Era	Core Technology	Energy Source	Employment Structure Change	Representative Case
1st (1760–1840)	Steam engine, power loom	Coal	Agriculture 75% → 22%; manufacturing rises to ~24%	Luddite movement
2nd (1870–1914)	Electricity, internal combustion engine	Oil & electricity	Manufacturing surge; new professional class emerges	Ford assembly line
3rd (1960–2000)	Computer, internet, semiconductors	Nuclear & renewables	Services 55% → 78%; knowledge workers rise	Silicon Valley
4th (2016–present)	AI, IoT, big data, robotics	Renewables + SMR	Knowledge work automation; AI specialists surge	ChatGPT (2022)

Graph 2-1: Improvement in Energy Efficiency



2.2 Energy Transition and Environmental Crisis

Flow of Energy Transitions

Before the industrial era, main energy sources were human and animal power. The first revolution shifted to coal, leveraging Britain's abundant reserves. The second brought petroleum to prominence, opening the era of internal combustion engines, automobiles, and aviation. In the 20th century, electricity became the dominant form, diversifying from coal-fired to hydro and nuclear power. In the 21st century, renewables have surged. Solar power costs fell from $0.38/kWh in 2010 to $0.04/kWh in 2023—a drop of over 90%—and in 2024, 92.5% of global new power capacity additions were renewable (IRENA Renewable Capacity Statistics 2025).

Such transitions dramatically improved energy conversion efficiency. Early steam engines achieved only 1–2% efficiency; Watt's improved version reached 3–5%; gasoline engines 20–30%; diesel engines 30–40%; and modern electric motors exceed 90%. From a life-cycle perspective, fossil fuel systems (extraction → refining → internal combustion) achieve only 15–20% final efficiency, while solar-electric systems (generation → ESS → EV) reach 75–85%.

New Challenges in the AI Era

The explosive growth of generative AI in the mid-2020s places unprecedented pressure on electricity demand. Training and inference of large models consume dozens of times more power than typical cloud computing. According to IEA Electricity 2026, data center power consumption is expected to rise from approximately 415 TWh in 2024 (1.5% of global electricity) to approximately 945 TWh by 2030 (~3%). In response, Google, Microsoft, Amazon, and other major technology firms are actively pursuing nuclear power purchase agreements (PPAs) and small modular reactors (SMRs), as nuclear can provide carbon-free, 24/7 baseload power.

The Reality of the Environmental Crisis

Energy transitions brought abundance, but fossil fuel dependence created grave environmental disasters. Coal use in the first revolution initiated widespread air pollution; the 1952 Great Smog of London killed approximately 12,000 people. The chemical industry of the second revolution caused water and soil contamination, exemplified by Japan's Minamata disease (mercury poisoning) in the 1950s–60s.

Climate change is the most serious consequence. Atmospheric CO_2 concentration rose from approximately 280 ppm before the industrial era to over 50% higher by 2026; global average temperature has increased approximately 1.3°C above pre-industrial levels (NOAA/Berkeley Earth, 2025). On current trends, a rise of +2.5–3°C by 2100 is projected (IPCC), with extreme weather, sea-level rise, and ecosystem collapse already occurring.

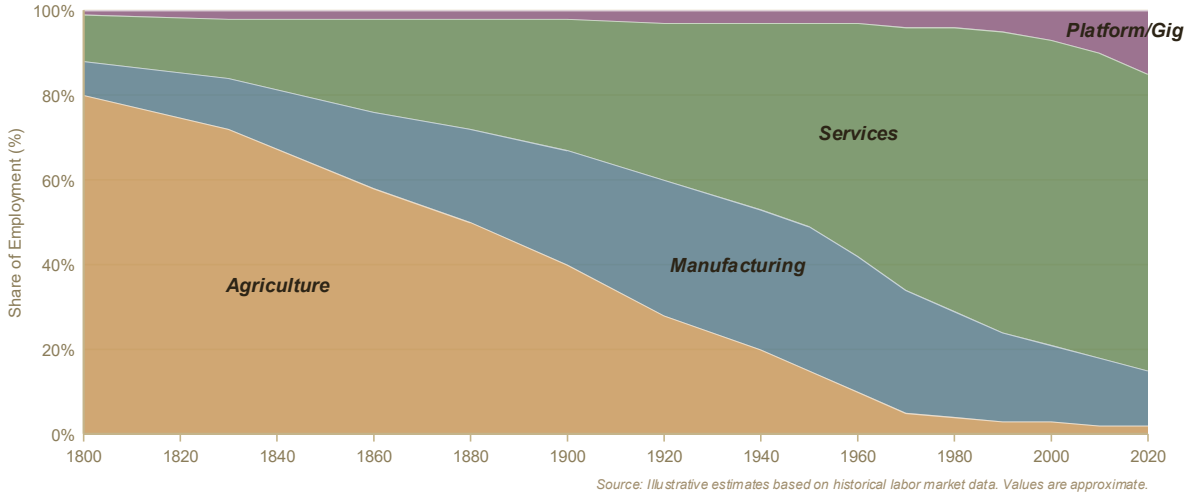
Industrial revolutions enriched humanity by changing energy sources, but at the cost of a climate crisis. Renewable energy, AI efficiency improvements, and carbon-neutral technologies now represent our last chance to overcome it.

3. Technology and Labor Market Changes

3.1 Evolution of Occupational Structure

Technological progress has fundamentally altered the nature of occupations. In agrarian societies, hereditary and lifelong single occupations were the norm; industrialization shattered this structure, and the information age established skilled knowledge work as the new standard. Today's platform economy is dismantling the traditional binary of "employer–employee" and forming a new labor paradigm centered on fragmented, task-based work.

Graph 3-1: Employment Shift by Sector



The explosive growth of the gig economy is evident worldwide. According to estimates from the World Bank and ILO, the global gig workforce in 2025–2026 ranges from 150 million to 450 million people, accounting for 4.4% to 12.5% of the total labor force. Platform-based labor markets are expanding rapidly, with particularly explosive demand for online gig work in developing countries.

Gig workers gain flexibility and autonomy but are placed in the blind spots of traditional employment safety nets. Platform algorithms determine task allocation, evaluation, and compensation, yet the criteria are largely opaque. Average working hours for Uber and DoorDash drivers in the US are around 14 hours per week (Gridwise data, 2025), and many workers multi-app to sustain livelihoods.

Safety issues are also serious. Uber's safety reports recorded 24,000 incident reports between 2017 and 2020, demonstrating that platform workers are constantly exposed to physical risks and accidents. This suggests that responsibility for worker safety is being shifted to individuals under the guise of "autonomy."

The gig economy provides freedom and opportunities, yet at the same time it is giving rise to new labor issues: income instability, inadequate welfare protections, and algorithmic exploitation. This shift is far more than a mere evolution in the forms of employment; it throws down an epochal challenge to completely redesign the rules and protective systems of future labor. But is the crisis unleashed by the disruptive innovations of the Fourth Industrial Revolution really limited to manual workers?

3.2 Crisis in White-Collar and High-Skilled Knowledge Work

While past industrial revolutions primarily replaced repetitive manual workers with machines, the AI-driven fourth industrial revolution is fiercely threatening even the high-education professional occupations that were once considered safe zones from technological unemployment. This is because large language models (LLMs) have acquired "cognitive proficiency" to analyze vast data, perform logical reasoning, and draft documents.

- Threat to entry barriers in law and accounting: Even the most conservative professions are being hit hard. AI now completes case law analysis and due diligence—core tasks of junior lawyers—in seconds, putting the very "entry pathway" of the professional job market at risk of extinction. Similarly, accountants face a sharp drop in demand for new hires as basic auditing and data classification tasks are automated.
- Downsizing in knowledge service industries: Major global consulting firms such as McKinsey & Co. have carried out large-scale layoffs in recent years, with further reductions planned. As data analysis and report writing—tasks once performed by high-level consultants—are streamlined by AI, organizational restructuring is accelerating.
- Crisis among big tech and SaaS companies: Entry-level software developers are also facing unprecedented employment difficulties. As AI reaches the stage of autonomously generating code and debugging errors, companies are shifting to structures where a small number of highly skilled developers lead AI tools, resulting in workforce reduction.

This phenomenon directly refutes the conventional social belief that "higher education guarantees safety from technological change." Employment stability is now determined not by the level of academic credentials but by the proportion of "non-routine creativity" and "complex judgment" in a role—that is, by how resistant the job is to algorithmic replacement.

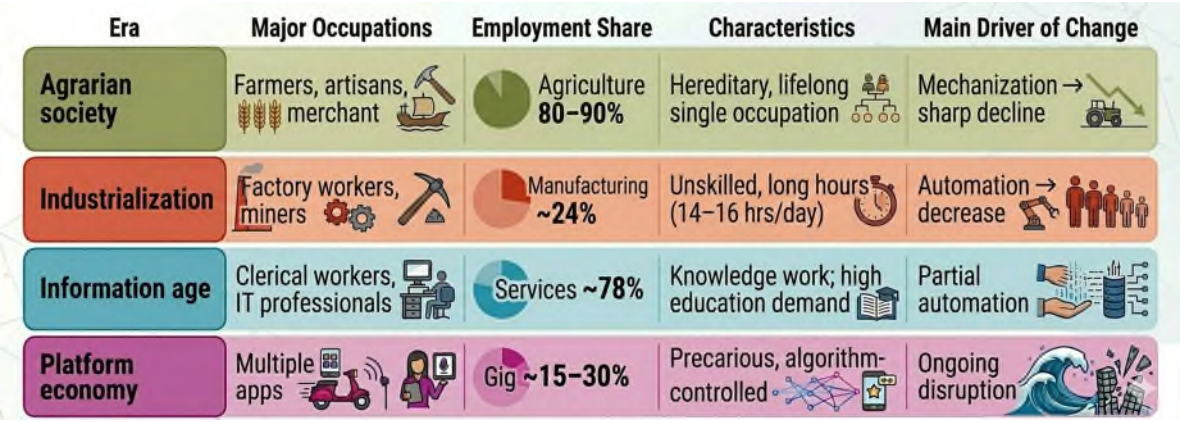


Figure 3-1: Major Occupational Changes by Era

3.3 The Bright and Dark Sides of the Platform Economy

Positive Aspects: Convenience and Efficiency

Platforms create value through network effects by connecting different user groups. According to JP Morgan analysis, Amazon revolutionized consumer convenience with one-click ordering and AI-based personalized recommendations; Amazon Prime members receive approximately $784 in annual benefits (5.6× the $139 subscription fee). Uber handles about 24 million trips per day across more than 10,000 cities in over 70 countries, becoming the representative model of the sharing economy. Platforms also provide small businesses with global market access and create new forms of employment through the gig economy.

Negative Aspects: Monopoly, Precarious Labor, and Collapse of Traditional Industries

The rapid growth of e-commerce platforms has dealt a heavy blow to traditional retail. Like the author's grandparents' Hanbok store in the traditional market, many markets and neighborhood commercial areas are rapidly declining; even some large offline stores are facing crises. In the US, the term "Amazonization" has emerged, with examples such as Borders (bankrupt 2011) and Toys "R" Us (bankrupt 2017). Former FTC Chair Lina Khan's 2017 paper pointed to Amazon's predatory pricing (initial losses → elimination of competitors → fee increases), vertical integration, and data monopoly as core problems.

The platform economy has expanded the gig economy but also increased precarious labor. Gig workers are classified as independent contractors, lacking employment stability and regular employee benefits. Algorithms decide task allocation, evaluation, and pay, but the criteria remain opaque.

Historical Resistance to Technological Innovation: Patterns and Modern Cases

When technology threatens jobs, workers have resisted throughout history. The Luddite movement of 19th-century England is the archetype, and the pattern persists today. In 2019, simultaneous strikes occurred at Amazon Prime Day in the US, Germany, and Poland over excessive work intensity (300+ packages per hour), algorithmic surveillance including bathroom time monitoring, and competition with robots. In 2023, the US West Coast port union threatened an 11-month strike against fully automated terminals, eventually settling with guarantees for current workers, retraining programs, and wage sharing of productivity gains. In 2025, the Korean Metal Workers' Union strongly opposed the pilot introduction of humanoid robots at Hyundai Motor's Ulsan plant, demanding employment guarantees and profit sharing.

Common patterns in historical resistance cases are as follows:

1. The core of resistance is not technology itself but the threat to survival rights from job and income loss.
2. Capital determines the timing and speed of technology introduction, while labor can only resist afterward—creating an asymmetry of bargaining power.
3. Governments initially support technology adoption but introduce labor protection policies belatedly when social conflict intensifies.

Successful transition examples—Germany's "Arbeiten 4.0" agreement (6-month advance notice, mandatory retraining, 40% profit sharing with workers)—demonstrate that prior consultation, profit sharing, and phased introduction are key. As a result, Germany's manufacturing robot density tripled from 2010 to 2020 while manufacturing employment actually increased by 5%.

3.4 Empirical Evidence of the Productivity Revolution

This section analyzes how much productivity has actually improved due to technological progress and its impact on employment, using empirical data.

Industry-Specific Productivity Gains

According to US Bureau of Labor Statistics data, US manufacturing output doubled from 1987 to 2019 while employment fell 33%—indicating roughly a threefold productivity increase. The same number of workers can now produce three times more, or one-third the workers can produce the previous output.

Agriculture shows the most dramatic gains. USDA data indicate that one US farmer fed 2.5 people in 1900, 15.5 in 1950, and 155 in 2000—a 62-fold productivity increase over 100 years, driven by tractors, chemical fertilizers, GM crops, and precision agriculture.

Ford's 1913 assembly line is the iconic symbol of the productivity revolution. Production surged from 10,000 to 2 million vehicles annually; significantly, lower prices from productivity gains exploded demand and actually increased employment (14,000 in 1913 → 100,000 in 1920). This shows that technology may displace specific jobs in the short term but can expand entire industries and create more jobs in the long run.

3.5 Productivity Revolution Empirical: AI Utilization Effect Experiment

To go beyond theoretical discussion, I designed and conducted an experiment during the preparation of this report to measure the actual impact of technology on productivity.

Experiment Design

Target task: Writing the entire report The Labor Paradox (~7,000 words, 6 chapters).
Participant: The author (high school student with limited prior academic writing experience).
Comparison method:

- A. Traditional method: Library literature search + handwritten notes + word processor
- B. AI-assisted method: Internet search + generative AI tools (Gemini, Claude, etc.)

Methodological note: Writing the full report twice separately was infeasible due to time and effort constraints. Therefore, part of Chapter 1 was measured using the traditional method, while Chapters 2–6 were written with AI assistance and timed. Quality evaluation combined self-assessment and advisor feedback.

A. Traditional Workflow

The biggest bottleneck was literature collection. As a high school student, access to academic databases was limited, and physical travel and manual organization consumed most time.

- Literature & data collection: 25–30 hours
- Outline & structure design: 5 hours
- Draft writing: 20–25 hours
- Review & revision: 8–10 hours
- Total estimated time: 58–70 hours (5.8–7 weeks at 10 hours/week)

B. AI-Assisted Workflow

Using AI as a collaborative partner dramatically reduced physical and cognitive load.

- AI-assisted research: 2 hours
- Structure design: 1.5 hours
- Draft writing: 0.5 hours
- Review & final editing & visualization: 8 hours
- Total measured time: 12 hours (1.2 weeks at 10 hours/week)

Note: *Time lost to trial-and-error with document editing and visualization AI tools was excluded from measurement, as this reflects a temporary learning curve that diminishes with practice.*

Table 3-2: Productivity Comparison — Time Investment by Stage

Stage	Traditional Method (est.)	AI-Assisted (measured)	Reduction Rate
Research & Data Collection	25–30 h	2 h	92–93%
Structure Design	5 h	1.5 h	70%
Draft Writing	20–25 h	0.5 h	98%
Review & Final Editing	8–10 h	8 h	~20%
Total	**58–70 h**	**12 h**	**~81%**

Average time savings: 81%. AI delivered the greatest efficiency gains in the research and drafting stages.

Table 3-3: Quality Comparison — Traditional vs. AI-Assisted Method

Quality Dimension	Traditional Method	AI-Assisted	Superior Method
Volume of Information	Medium	High	AI-Assisted
Depth of Information	Medium	High	AI-Assisted
Recency / Up-to-dateness	Low	High	AI-Assisted
Diversity of Perspectives	Limited	Rich	AI-Assisted
Original Argumentation	High	Medium	Traditional
Sentence Polish / Completion	Medium	High	AI-Assisted
Consistency & Conciseness	Medium	High	AI-Assisted
Data Visualization	Low	Very High	AI-Assisted

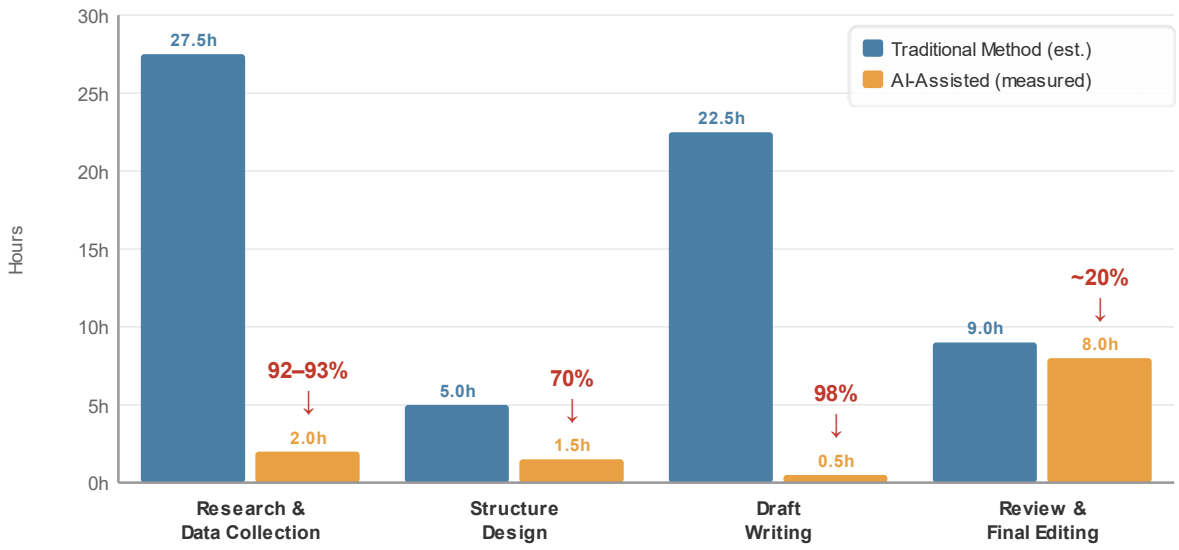
AI excelled in volume of information, depth, recency, diverse perspectives, sentence polish, consistency and data visualization. The traditional method was stronger in original argumentation and personal voice.

Limitations and Risks

- Traditional method: Time inefficiency, limited access to the latest materials, and access barriers for high school students.
- AI method: Hallucination, source errors, plagiarism risk, and over-dependence.

Countermeasures adopted: manual verification of 10–15 key statistics and citations, rewriting AI-generated text, and mandatory addition of personal perspectives (e.g., grandparents' interview).

Graph 3-2: AI-Assisted vs. Traditional Productivity Comparison (Author's Experiment)



The experiment clearly demonstrated that generative AI can improve knowledge-work productivity by an average of 81%—meaning roughly 5× more output in the same time. However, of the 12 hours, AI-led time was approximately 7 hours (58%) while human judgment, verification, and editing took approximately 5 hours (42%). Setting direction, fact-checking, and adding original perspective remain irreplaceable human roles.

The most important lesson: future knowledge workers need not mere technical skill but AI literacy—the ability to effectively direct, verify, and augment AI with their own thinking. AI is a tool; how it is used remains a human decision.

Productivity–Employment Paradox

Historically, productivity gains have increased employment in the long run. ILO data show global labor productivity rose 8× from 1950 to 2020, while total employment grew 3×. Yet the current AI revolution differs in speed, scope, and substitutability. McKinsey (2023) projects that by 2030, 800 million jobs worldwide may need transition, but 850 million new jobs could be created. The key lies in the speed of transition and retraining infrastructure.

"AI-assisted 12 hours vs. non-AI 64 hours = 5.3× productivity gap." What should we do in the future? It has become evident that AI is no longer a matter of choice. However, what to create—and what meaning to give to creation—still remains firmly in the hands of humans. This deeper question will be explored further in Chapters 5 and 6.

4. Future Scenarios

Chapters 1 through 3 examined the past and present: how technology has transformed occupations, how workers have resisted, and how productivity gains have historically increased employment in the long term. But does "the past guarantee the future"? We must be cautious. The ongoing AI and robotics revolution differs from previous ones in speed, scope, and depth.

This chapter presents two scenarios. The positive scenario depicts technology leading humanity to unprecedented abundance and freedom. The negative scenario portrays deepening polarization and chaos. Which path becomes reality has not yet been decided.

4.1 Stages of Transition: 2025–2070

Short-Term Future (2025–2035): The Beginning of Transition

By the 2030s, major cities in advanced countries will have transformed into smart cities. Autonomous buses and taxis will fill the roads, and AI will optimize traffic signals in real time, reducing congestion by 50%. Singapore already uses its Smart Nation project to install sensors citywide, optimizing energy use, waste collection, and public safety with AI.

With full self-driving commercialization, traffic accidents are expected to drop by 90%. According to the US National Highway Traffic Safety Administration (NHTSA), 94% of crashes involve human error (drunk driving, drowsiness, distraction, etc.). Autonomous systems maintain 24/7 attention, 360° sensing, and millisecond reaction times. McKinsey estimates that widespread Level 5 autonomy could reduce annual global traffic fatalities from 1.35 million to around 0.13 million.

Medical AI will surpass humans in specific domains. Google DeepMind researchers behind AlphaFold won the 2024 Nobel Prize in Chemistry for protein structure prediction. Imaging AI already diagnoses lung and breast cancer more accurately than human radiologists. In 2030s hospitals, AI will provide first-line diagnosis, telemedicine will become standard, personalized treatments will emerge from genetic analysis combined with AI, and new drug development timelines will shrink from 10 years to 3 years.

Education will undergo revolution. Tools like Khan Academy's Khanmigo and Duolingo's AI tutors already offer personalized learning. By 2030, AI personal tutors will adapt curricula to each student's pace, comprehension, and interests, provide real-time feedback, and enable access to global lectures in native languages via real-time translation—narrowing educational gaps.

Shorter working hours will become widespread. Successful trials in Iceland (2015–2019), Belgium (2022), and the UK (2022) showed maintained or improved productivity with better well-being. In the UK trial (61 companies, 2,900 employees), 92% of firms decided to continue four-day weeks; revenue rose 1.4%, turnover fell 57%, and sick days dropped 65%. By 2030, approximately 30% of OECD countries are expected to legislate four-day or 35-hour workweeks.

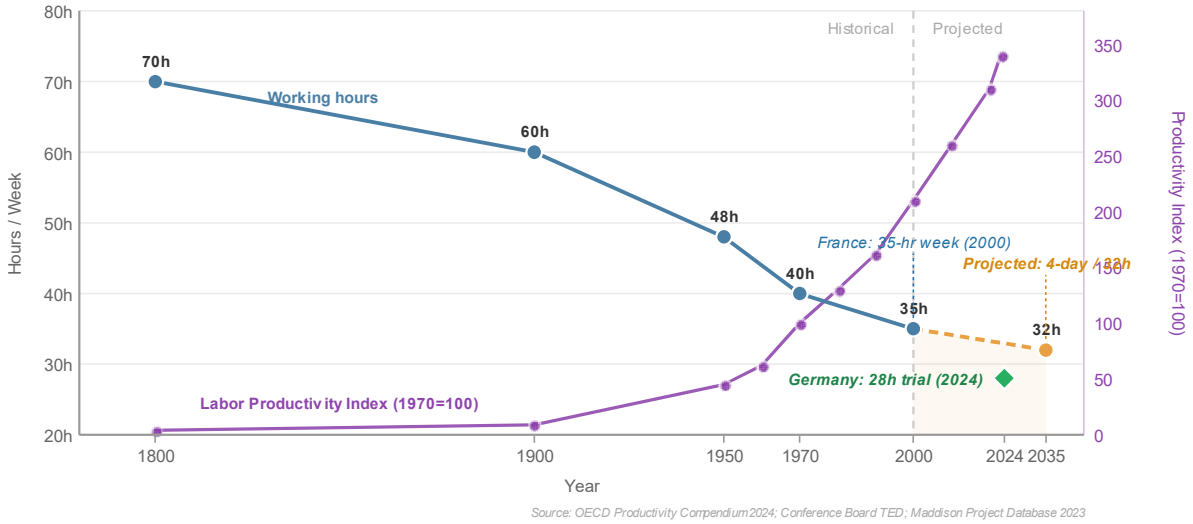
The 3D jobs (Difficult, Dirty, Dangerous) will gradually diminish. Construction robots handle bricklaying, welding, and painting; mining uses unmanned excavators and remote-controlled trucks; logistics robots manage heavy lifting and high-altitude work. Hyundai Engineering & Construction's 2023 AI-based safety system already reduced accidents by 30%.

South Korea's unique situation deserves attention. Holding both the world's lowest fertility rate and the highest industrial robot density is no coincidence. In the positive scenario, this is perfect matching: robots fill labor shortages, replace avoided 3D jobs, and compensate for productivity as the youth share declines due to aging. South Korea could become a global "automation test-bed" model for adapting to population structural change.

Mid-Term Future (2035–2045): Acceleration of Transition

Productivity surges will lead to legislated shorter working hours—four-day weeks or 35-hour standards becoming common in major OECD countries. Historically, working hours have continuously declined: 70 hours/week in the 1800s → 60 in the 1900s → 48 in the 1950s → 40 in the 1970s. France introduced 35 hours in 2000; parts of Germany are experimenting with 28 hours.

Graph 4-1: Working Hours vs. Labor Productivity Cross-Plot



Leisure industries will explode. Three-day weekends will boost domestic and international travel, cooking, music, art, sports, and hobby education; metaverse, VR/AR content consumption will skyrocket; yoga, meditation, and fitness/wellness sectors will expand dramatically.

Carbon neutrality will near completion. By 2040, power, transport, buildings, and industry will achieve full transition. Renewables will be far cheaper than fossil fuels; energy storage systems (ESS) will solve intermittency; nuclear fusion will enter commercialization.

Long-Term Future (2045–2070): New Social Contract

The technological singularity debate may become reality (Ray Kurzweil predicts approximately 2045). Even without full singularity, progressive automation will continue.

By the 2050s, fully automated economies will appear. Unmanned farms handle seeding to harvesting, processing, and packaging. Dark factories (lights-out, fully unmanned) operate 24/7; humans never need to enter. Autonomous trucks load goods at factories; drones and delivery robots bring them to homes. Robots install and maintain solar panels and wind turbines; AI optimizes real-time power supply.

Jeremy Rifkin's "zero marginal cost society" may materialize. As the marginal cost of production approaches zero, prices approach zero. Digital goods already have near-zero marginal cost; physical goods may follow with AI and robotics.

In a 2050 price imagination: food (unmanned farms + cultured meat → 90% cost reduction), manufactured goods (automation → 95% reduction), energy (fusion + renewables → electricity 1/10th price), housing (robotics/3D-printed construction → 95% reduction), education (free AI tutors), healthcare (AI diagnostics + preventive medicine → 70% reduction).

Full Universal Basic Income (UBI) becomes feasible. With sufficient automation, governments could collect 30–40% of GDP via robot taxes, wealth taxes, land taxes, carbon taxes, etc., to provide UBI to all citizens at approximately 40–50% of median income—enough for basic living, with optional work for higher quality of life.

Labor itself will be redefined. The capitalist notion "work = means of survival" will fade. Work becomes a means of self-realization, social contribution, and meaning creation. Unpaid but valuable activities (childcare, caregiving, arts, volunteering, environmental protection) gain recognition. Time banks or alternative currencies may emerge.

Medical revolution and life extension also materialize: CRISPR eliminates genetic diseases, nanobots remove cancer cells and prevent clots, 3D-printed organ regeneration, and telomere

extension and cellular senescence inhibition. Harvard geneticist George Church estimates an average lifespan of 100 by 2050 and 150 by 2100 is possible.

Negative Scenario: Polarization and Chaos

Technology poorly managed can lead to dystopia.

In the short term: mass unemployment and social unrest if AI and robots displace jobs faster than new jobs and retraining can respond. McKinsey (2023) warns that full self-driving trucks by 2028 could put approximately 1.7 million US truck drivers at risk; Gartner (2024) predicts AI chat agents will replace 60% of call-center work by 2029.

Platform monopolies intensify. Amazon, Google, and Meta solidify dominance via data and AI advantages; new entrants become nearly impossible due to massive data requirements for training; predatory pricing, unilateral fee hikes, and vendor lock-in worsen.

In the long term: extreme polarization. By 2070, society could split between an ultra-wealthy 1% (AI and robot company owners controlling 50% of global wealth, lifespan 150) and a 99% underclass (barely surviving on UBI, jobless, escaping into VR, average lifespan 80). The gap may become so vast that they no longer seem the same species.

AI dictatorship is possible. Social credit systems expand globally; AI monitors every action (CCTV, smartphones, social media); low scores block loans, jobs, and travel; criticizing government lowers scores. Orwell's 1984 becomes reality.

Environmental catastrophe remains possible. If technology fails to save the environment, +3°C by 2100 and a 2m sea-level rise would flood Bangladesh, the Maldives, New York, and Shanghai; food production would drop 30%, causing mass starvation; 50% species extinction would collapse ecosystems; and 1 billion climate refugees could trigger border conflicts and resource wars.

Graph 4-2: Positive/Negative Scenario Matrix

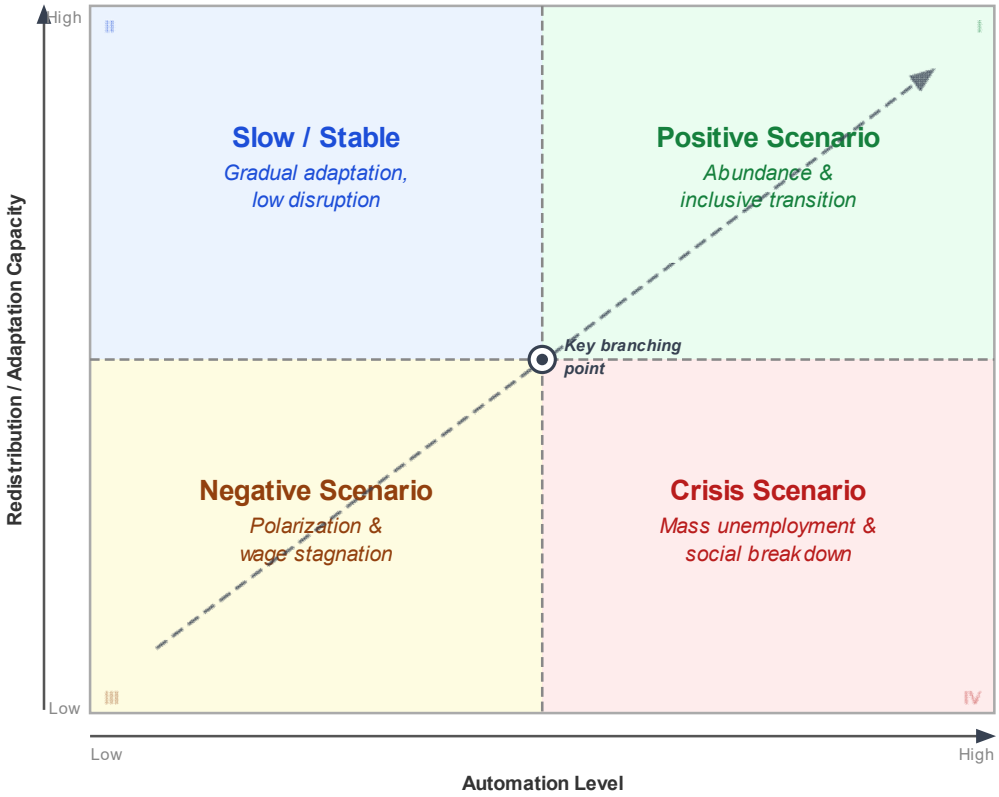
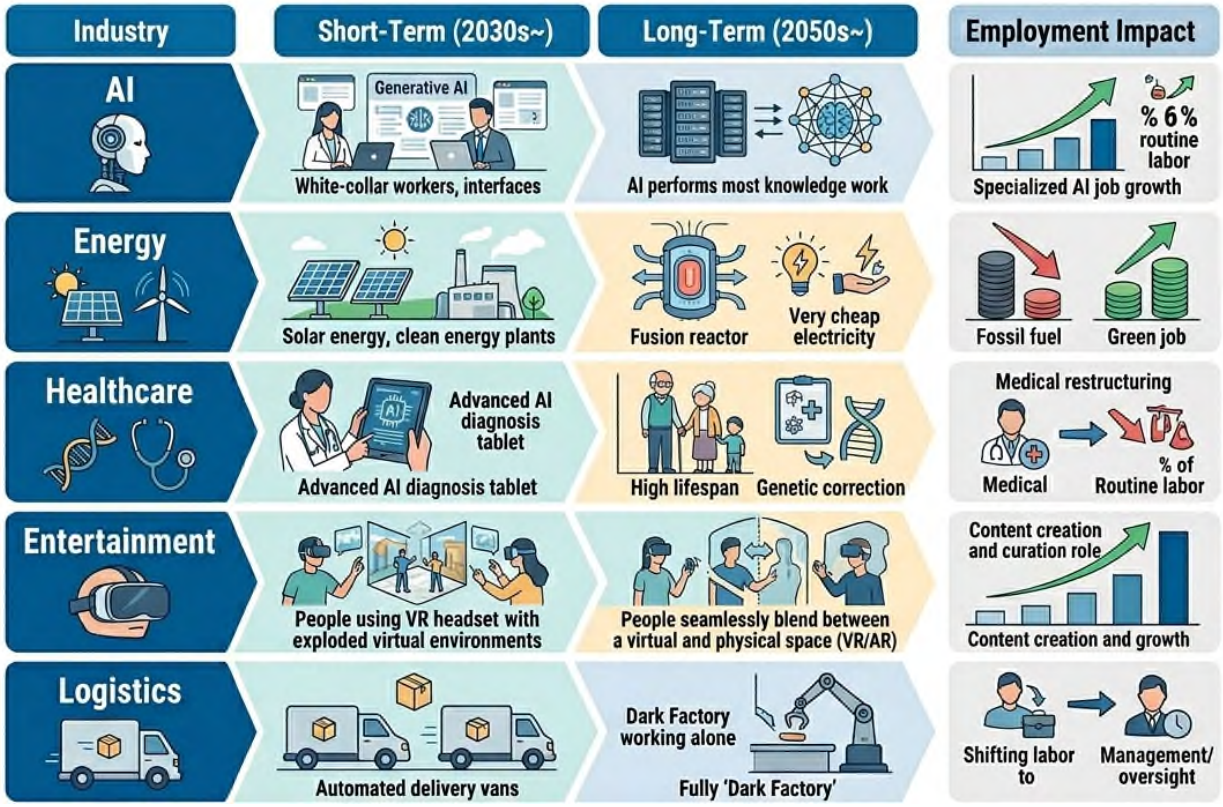


Table 4-1: Summary of Future Outlook by Industry

Industry	Short-Term (2030s~)	Long-Term (2050s~)	Employment Impact
AI / Robotics	Widespread corporate adoption; generative AI in white-collar roles	AI performs most knowledge work; specialized AI roles surge	White-collar restructuring; surge in AI specialists
Energy	Renewables become mainstream; fossil fuel phase-out begins	Fusion commercialization; electricity cost falls to 1/10th	Fossil fuel jobs decline; green jobs increase substantially
Healthcare / Biotech	AI diagnostics become standard; telemedicine widespread	Average lifespan ~100; genetic diseases largely eliminated	Medical workforce restructuring; precision medicine specialists
Entertainment	Metaverse economy takes off; VR/AR content explodes	Virtual and real boundaries blur; immersive experiences dominate	Explosive growth in content creation and curation roles
Manufacturing / Logistics	Autonomous driving and robotic logistics commercialize	Dark factories (fully unmanned) operate 24/7	Routine labor replaced; human roles shift to management/oversight



4.2 Key Challenges and Risks

Technological Deflation and Economic Restructuring

Price collapse from technology (technological deflation) differs from traditional deflation (e.g., Japan's lost decades). Supply explosion lowers prices, benefiting consumers, but exposes GDP measurement limitations. Alternative indicators—Genuine Progress Indicator (GPI), Bhutan's Gross National Happiness (GNH), and energy consumption metrics—may better capture societal well-being in this era.

Extreme Wealth Polarization

As automation advances, returns to capital overwhelm returns to labor. Robot and AI owners accumulate vast wealth while workers lose jobs. US top 1% wealth share rose from 23% in 1989 to 32% today; acceleration could cause severe social consequences. Potential remedies include UBI, wealth taxes, and robot taxes.

Ethical Issues

AI reproduces data biases and discrimination (Amazon hiring AI discriminated against women, 2018; COMPAS recidivism AI was biased against Black defendants; facial recognition achieves lower accuracy for Black women). Solutions include diverse training data, explainable AI, and regular bias audits.

Bioethics challenges arise: gene-edited babies (He Jiankui, 2018), brain-computer interfaces (Neuralink), and artificial wombs (successful sheep experiments, 2017) all raise the question: "What does it mean to be human?"

Human dignity is threatened. If robots do everything better, are humans worthless? In capitalist societies, "usefulness" equals value. Without a cultural shift, many people in automated societies may feel valueless.

Environment and Resource Issues

Technology could save or destroy the environment. Positive outcomes include: zero-carbon renewables, 90% air pollution reduction via EVs, 50% fertilizer and pesticide cut via precision farming, 90% methane reduction via cultured meat, and zero-waste circular economy adoption.

Negative outcomes include: AI power demand surges (IEA 2024: data centers from 1–1.5% to 3–4% of global electricity by 2030); rare-earth mining (for EV batteries, solar panels) causing massive soil and water pollution; and short replacement cycles of AI chips and robots generating growing e-waste.

Geopolitical Conflict and Global Inequality

Nations dominating AI, semiconductors, and quantum computing will dominate the 21st century. US–China tech competition intensifies (chip export bans, AI competition, 5G/6G alliances). Resource conflicts over lithium, rare earths, and cobalt remain possible.

Lethal autonomous weapons (LAWS) raise concerns: no human oversight in target selection, lowered threshold for armed conflict, and hacking risks. Over 30 countries discussed regulation in 2024, but the US, Russia, and China remain reluctant.

Digital divide widens: most African countries have less than 30% internet penetration. AI education and robotics infrastructure remain accessible only to advanced nations—posing a "digital colonialism" risk. UN SDGs emphasize "leave no one behind"; technology transfer, knowledge sharing, and fair trade are essential.

5. Roles for a Sustainable Future

Chapter 4 presented two futures: a utopia of abundance and freedom, or a dystopia of polarization and chaos. The path depends on choices made by governments, corporations, and individuals. This chapter clarifies each stakeholder's role. Harmony among them can avoid dystopia and move toward a sustainable future.

5.1 Government: Institutions and Safety Nets

Education Innovation: Toward a Lifelong Learning Society

The most critical government role is education reform and safety nets to help all citizens adapt to rapid change. The current education system was designed to meet the needs of the 19th-century industrial era. Standardized testing and memorization-centered education were well suited to producing factory workers, but they are no longer effective in the AI era.

Finland's Phenomenon-Based Learning crosses subject boundaries, focuses on projects, evaluates via portfolios, and treats teachers as facilitators—maintaining top PISA rankings while achieving world-leading student happiness.

Singapore's SkillsFuture provides every citizen over 25 with S$500 credit, up to 90% subsidies, and tax benefits for corporate retraining; over 60% of the workforce has participated since 2015.

Requirements for the future:

- Mandatory AI literacy from elementary school (programming, data analysis, AI ethics, prompt engineering)
- Breaking down university departmental walls and strengthening interdisciplinary fusion
- Lifelong learning accounts for all citizens, with expanded industry-linked practical education
- Shifting evaluation from "correct answers" to "what questions did you ask AI?" and "how did you critically verify AI outputs?"

Social Safety Nets: Combining Flexibility and Security

Denmark's flexicurity model achieves both labor market flexibility and strong safety nets. Companies can hire and fire easily, but the unemployed receive 70–80% of their previous salary for up to 2 years with mandatory retraining—resulting in less than 5% unemployment and top-tier national happiness.

Sweden's Job Security Councils connect laid-off workers to intensive 6-month support (résumé, interview, retraining), achieving 85% re-employment.

To cover platform workers and freelancers, France's universal unemployment insurance (2019) reclassifies "economically dependent workers" for social insurance coverage.

Basic Income and Working-Time Reduction Experiments

UBI pilots show positive signals. Finland (2017–2018) gave 2,000 people €560/month unconditionally—employment held steady while mental health and happiness improved significantly. Kenya's long-term experiment (GiveDirectly, 2016–present) reduced poverty, domestic violence, and improved child nutrition.

Four-day week trials succeeded (Iceland public sector: productivity maintained or improved; UK: 92% continuation with revenue +1.4%, turnover -57%, sick days -65%).

Government actions:

- Regional and age-group UBI pilot programs
- Expand four-day week trials toward legislation by 2030
- Secure funding via carbon tax, robot tax, and wealth tax

Technology Regulation and Ethics

The EU AI Act (2024) is the world's first comprehensive framework—risk-based four-tier classification:

- Unacceptable risk (social credit, real-time biometric surveillance) → banned
- High risk (recruitment, credit scoring, medical, judicial) → strict pre-approval and transparency requirements
- Limited risk (chatbots) → disclosure obligation
- Minimal risk → unregulated

GDPR sets the global personal data protection standard: explicit consent, right to be forgotten, data portability, fines up to 4% of revenue or €20M.

Government's future role is not just welfare provider but facilitator that cushions technological shocks and enables universal participation in the new era.

Table 5-1: Key Government Tasks and Best Practices

Domain	Core Task	Best Practice	Improvement Direction
Education	Lifelong learning system	Singapore SkillsFuture	Expand credits; mandate AI literacy from elementary school
Safety Net	Flexicurity model	Denmark flexicurity (70–80% benefits + mandatory retraining)	Extend coverage to platform and gig workers
Income Support	Basic income experiments	Finland / Kenya UBI pilots	Launch regional and age-group pilots
Working Hours	Working-time reduction	Iceland four-day week (2015–2019)	Legislate 35-hour / 4-day standard by 2030
Regulation	AI ethics legislation	EU AI Act (2024)	Mandate algorithm transparency and bias auditing

5.2 Corporations: Innovation and Responsibility

Corporations stand at the forefront of technological innovation but must ensure it benefits society as a whole. Beyond profit maximization, creating shared economic and social value will become the core competitiveness of future companies.

Creating Shared Value (CSV)

Michael Porter's CSV integrates solving social problems with profit generation. Unilever's Sustainable Living Plan (2010–) targeted improving health and hygiene for 1 billion people, halving environmental footprint, and creating 1 million women's jobs by 2030—sustainable brands (Dove, Ben & Jerry's) grew 69% faster than average, reduced water use by 32%, CO_2 by 52%, and achieved €51B revenue in 2020.

Just Transition

Automation is inevitable, but workers must not be abandoned. Shell's Just Transition Fund (US$1B) supports oil-to-renewables workers with retraining, redeployment, and early retirement compensation. Germany's coal phase-out (Kohleausstieg) invests €40B by 2038 for miner retraining, regional diversification, and compensation.

Corporations should adopt: minimum 6-month advance notice, union consultation, employment guarantee plans, and sharing a portion of productivity gains with workers.

ESG and Responsible Investment

ESG is now central to investment decisions. BlackRock's CEO Larry Fink (2020) called climate

change the biggest investment risk. Tesla excels in E (EVs), S (autonomous safety), G (transparency) → approximately $1.5T market cap (2025). Ørsted transitioned from coal to the world's #1 offshore wind operator → 10× market cap increase, named most sustainable company (Corporate Knights 2023).

Ethical AI Development

Google AI Principles (2018) emphasize beneficial, fair, safe, and transparent AI. Microsoft's Responsible AI includes ethics boards, Fairlearn open-source bias tool, and developer training. OpenAI's original "safe AGI" mission faced tension after the 2019 for-profit shift and 2024 speed-versus-safety debates.

Corporations must match innovation speed with responsibility and ethics.

Table 5-2: Key Corporate Tasks and Best Practices

Domain	Core Task	Best Practice	Expected Effect
CSV	Integrate social and economic value creation	Unilever Sustainable Living Plan	Sustainable brands grew 69% faster than average
Just Transition	Worker retraining and fair compensation	Shell Just Transition Fund (US$1B)	Reduced social conflict; improved corporate reputation
ESG	Environment, Society, and Governance improvement	Tesla EVs; Ørsted offshore wind	Attract long-term investment; enhance sustainability
AI Ethics	Bias removal and algorithmic transparency	Microsoft Fairlearn	Build stakeholder trust; reduce regulatory risk
Work Model	Flexible and four-day work schedules	Basecamp remote-first model	Maintain productivity; increase employee satisfaction

5.3 Individuals: Adaptation and Growth

Lifelong Learning and Adaptability

In the future labor market, the most important competency is learning ability. The World Economic Forum's Future of Jobs Report 2023 identified the following ten core skills for the future labor market: analytical thinking, creative thinking, technological literacy (AI and big data), curiosity and lifelong learning, resilience and flexibility, empathy and active listening, leadership and social influence, quality control, systems thinking, and service orientation and customer service.

Online platforms such as LinkedIn Learning, Coursera, edX, MIT OpenCourseWare, and Khan Academy provide access to lectures from the world's top universities at low cost or for free. MIT OpenCourseWare offers more than 2,500 courses completely free of charge, while Khan Academy delivers free education from elementary school to adult level. Individuals must actively seize these educational opportunities. Each year, learn one new technology, invest time and effort in useful online lectures and information, and continuously develop the ability to think critically and reason deeply.

Critical Thinking and AI Literacy

In an era of information overload, the key is critically evaluating information. Identify deepfakes and fake news, verify sources (Snopes, FactCheck.org), interpret statistics (correlation ≠ causation), and understand AI limitations (hallucination, bias). Curate valuable and trustworthy information amid "information pollution."

Always verify AI outputs, use AI as an assistant (not a replacement), make final judgments yourself, and consider ethical use (plagiarism, copyright, privacy).

Creativity and Uniquely Human Capabilities

AI struggles with true creativity, empathy, and ethical judgment. IDEO and frog design emphasize design thinking: empathy, problem definition, ideation, prototyping, and testing—human-centered thinking that is difficult for AI to replicate.

Build creativity through diverse experiences (travel, reading, arts, sports), interdisciplinary exchange, embracing failure, and sustained curiosity.

Pursuing a Meaningful Life

When work is no longer a means of survival, what is life for? Viktor Frankl wrote that "meaning is most important to humans." In concentration camps, those with reasons to live survived.

Clarify personal values, contribute to society (volunteering, mentoring, environmental protection), invest in relationships (family and friends), and enjoy leisure.

Sustainable Consumption and Citizenship

Individual consumption shapes corporate behavior. Patagonia's "Don't Buy This Jacket" campaign paradoxically increased loyalty. TOMS' "One for One" model demonstrated the power of values-aligned purchasing.

Practice minimalism, choose high-ESG companies, recycle and reuse, prioritize local products

(reducing carbon footprint), stay engaged with environmental issues, vote, sign petitions, support NGOs, speak on social media, and engage in local communities.

Table 5-3: Integrated Key Roles by Stakeholder

Stakeholder	Core Role	Main Tasks	Practical Methods	Expected Outcome
Government	Facilitator	Education innovation; safety nets; income support	Lifelong learning accounts; UBI pilots; AI ethics legislation	Social stability and fair transition
Corporation	Innovator	CSV; Just Transition; ESG; ethical AI	Retraining investment; carbon reduction; responsible AI development	Sustainable growth and shared prosperity
Individual	Agent	Lifelong learning; creativity; meaningful contribution	AI literacy; critical thinking; community and social engagement	Enhanced adaptability and personal fulfillment

6. Conclusion

6.1 Research Summary

This study began with the paradox of "job destruction and labor shortages coexisting." Analysis revealed it is not a true paradox. Technology destroys jobs while simultaneously creating them. The real issue is friction during transition: temporal mismatch (short-term loss, long-term gain), spatial mismatch (specific regions and industries), and skills mismatch (disappearing vs. emerging jobs).

History shows technological progress increases long-term employment. Global labor productivity rose 8 times from 1950 to 2020, while total employment grew 3 times. Yet the AI revolution differs in speed, scope, and depth—replacing both manual and knowledge work comprehensively.

Low birth rates and aging present challenges, but they are not necessarily catastrophic. Korea, holding both the world's lowest fertility rate and the highest robot density, can serve as an "automation test-bed," compensating for labor shortages through technology. Even if the population declines, if per-capita income rises, the environment improves, leisure time increases, and people can enjoy a more affluent and comfortable life, can we truly define this phenomenon as a problem of "population extinction"?

6.2 Policy Recommendations

Short-Term (2025–2030)

1. Mandatory AI literacy from elementary school
2. Introduce Danish-style flexicurity (70–80% unemployment benefits + mandatory retraining)
3. Establish risk-based AI regulation framework referencing the EU AI Act
4. Launch regional and age-specific UBI pilot programs
5. Expand the four major social insurances to cover platform workers

Mid-Term (2030–2040)

1. Legislate a four-day week or 35-hour standard
2. Expand universal basic services (UBS) in healthcare, education, and housing
3. Achieve carbon neutrality by 2040
4. Strengthen international cooperation for technology transfer to developing countries

Long-Term (2040–2070)

1. Realize full UBI funded by robot and wealth taxes
2. Redefine labor as a paradigm for self-realization
3. Expand participatory and deliberative democracy and establish AI governance

6.3 Looking Forward

This report began with the author's grandparents' Hanbok store in a traditional market—a community hub where customers received personalized fabric recommendations and shared holiday meanings. Yet platform algorithms instantly turned decades of "experiential value" into a "price-competitiveness disadvantage."

Is technological progress truly creating better lives, or merely destroying many livelihoods for the efficiency of a few?

The "Labor Paradox" this research uncovers offers a new direction. Technology itself did not destroy my grandparents' store—rather, the proper use of technology holds the key to resolving such crises.

Ultimately, future labor should not be a struggle for survival under giant capital and technology, but a journey of self-realization using technology as a tool to express individuality and values. I

hope this report serves as small groundwork for designing a future where technology enhances human dignity and protects even the value of small neighborhood shops.

Limitations of the Study

1. Relied mostly on secondary sources; failed to secure field interviews (e.g., delivery riders)
2. Advanced-country centered; lacking developing-country perspectives
3. Qualitative focus without quantitative modeling
4. Chapter 4 scenarios are possibility spectra, not deterministic predictions
5. Extensive AI use; all facts were verified but errors remain possible

Preparing for Future Occupations

The most significant insight derived from this study is that future society will reorganize around three fundamental axes.

1. AI and robotics as core industrial drivers — almost every sector (manufacturing, logistics, construction, agriculture) will automate; understanding robot physics, AI learning mechanisms, and human-machine collaboration becomes essential. This is why I pay attention to mechanical engineering.
2. Explosive growth of the entertainment industry — four-day weeks and UBI will increase leisure time dramatically; play, culture, arts, metaverse/VR/AR, games, film, music, and sports will become life's center. Even with AI-generated content, people crave "human" stories and emotion. This intersection—technology meeting human creativity—is where new value emerges, drawing my interest in entertainment development.
3. Fundamental economic system transition — UBI, carbon and robot taxes, and UBS will shift redistribution structures; the "no work, no eat" labor ethic will collapse. The core question becomes "how to fairly distribute wealth"; happiness and sustainability indicators will overtake GDP. Understanding the technological impact on inequality and new models requires economics.

Fusing these three—mechanical engineering (the foundation of technology), entertainment (the fulfillment of human desires), and economics (the framework for social structure)—makes it possible to design a better future.

Insights from this research point to a concrete path forward. These understandings of future society extend beyond individual career planning and can provide meaningful direction for peers

of the same generation. We all stand before the question, "What occupation should I choose?" Yet far more important is the question, "What values will I pursue in life?"

Short-Term (University, 4 Years): Building an Interdisciplinary Foundation

University offers the opportunity to integrate the three axes of mechanical engineering, entertainment development, and economics. Mechanical engineering provides a solid grasp of the physical foundations of robotics; an entertainment major enables exploration of AI combined with creativity, the metaverse, and game design; economics offers insight into macro-level social change and policy contexts.

At the same time, writing skills will be continuously refined—not toward becoming merely a technically proficient engineer, but toward becoming a communicator capable of conveying the societal meaning of technology to the wider world. The four years of university are not simply a time for acquiring knowledge; they represent a critical period for cultivating a mindset that bridges disparate fields.

Mid-Term (10 Years After Graduation): Gaining Field Experience and Building Networks

The decade following graduation serves as a period for accumulating practical experience. This can involve working at the technological forefront in AI and robotics at major technology companies, participating in global macro-policy initiatives at international organizations (UN, OECD, ILO), or conducting research and policy recommendations at think tanks (Brookings Institution, RAND Corporation).

The core focus of this phase lies in understanding the complexities of real-world settings. It allows for a deep grasp of how theories learned in university function—or fail to function—in actual practice. At the same time, a global network is built. Since the future is not the problem of a single nation but a shared human challenge, relationships with people from diverse countries and fields become essential.

Long-Term: Serving as a Bridge Between Technology and Policy

Over the long term, the aspiration is to occupy the role of both technical expert and policy advisor—contributing to society not from within the ivory tower of academia alone, nor from the narrow confines of practical fieldwork, but from the space between: a bridge connecting scholarly depth with real-world impact. Rigorous inquiry finds expression in academic research, while the same ideas are translated into accessible form for the broader public.

Coexistence of Technology and Humanity

The most important realization reached at the end of this study is that the situation is "not a paradox, but a transition." Humanity has always undergone technological shifts and adapted to them. However, the current wave of AI and automation differs in speed, scope, and depth from anything that came before. It replaces both physical labor and knowledge work, impacting routine tasks as well as creative ones.

Two futures remain possible: a dystopia in which technology is monopolized by a few, leading to extreme inequality, or a utopia in which the benefits of technology are shared, bringing prosperity and freedom. The future depends on choices that are made. Optimism is warranted. Humanity, though imperfect, learns and adapts. It has overcome the Black Death, world wars, and the Cold War. The AI era will also be navigated with wisdom.

What Does the Future Hold?

A complete answer has not yet been found—and perhaps never will be. The future is uncertain, life is unpredictable, and plans are always subject to change. Nevertheless, the direction remains clear.

The Luddite movement ended in failure, yet it carried profound meaning. The pain and struggle of those workers gave birth to labor rights, the welfare state, and social safety nets. History does not move in a straight line; it spirals. It circles, yet always ascends.

The author's grandparents' market no longer exists. Yet through their generation, our parents', now our own—we prepare to create a new future.

We stand at a turning point. AI and robots represent the most powerful tools humanity has ever created. For the first time in history, humanity encounters a species superior to our own in capability. Whether this turns out to be a blessing or a curse depends entirely on us.

A future where technology and humanity collaborate, where growth and sustainability coexist, where efficiency and humaneness are in balance—this is possible. It can be realized if we build it together.

The Labor Paradox is not a paradox; it is an invitation sent to humanity:

to choose the future we desire.

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